INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

CASTING POWDER, PROPELLANT

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of casting powder for cast double base propellants.

1.2 Classification. Casting powder covered by this specification are of the following types, as specified (see 6.2):

1.2.1 Types.

a. Type I – Nitrocellulose to nitroglycerin ratio of 4.394.

b. Type II – Nitrocellulose to nitroglycerin ratio of 4.094.

c Type III – Nitrocellulose to nitroglycerin ratio of 4.786.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification, recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-G-155	-	Graphite, Dry (For Use in Ammunition)
MIL-DTL-244	-	Nitrocellulose
MIL-N-246	-	Nitroglycerine
MIL-T-301	-	Triacetin (Glyceryl Triacetate)
MIL-A-2550	-	Ammunition, General Specification for

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-N-3399	-	2-Nitrodiphenylamine

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-286	-	Propellants, Solid: Sampling, Examination, and Testing
MIL-STD-1171	-	Acceptance and Description Sheets (for Propellants and Explosives)

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND PUBLICATIONS

WS 23141	-	Carbon Black (Powdered)
OD 17118	-	Preparation of Casting Solvent for Solid Propellant
OD 19477	-	Density Determination, Mercury Method, Casting Powder
OD 19554	-	Determination of Burning Rate Versus Pressure from Propellant Test Slabs
OD 19555	-	Manufacture of Solid Propellant Test Grain
OD 24654	-	Evaluation Grain, 4.5-Inch Solid
OD 25216	-	Screen Loading Density, Determination of
OD 26254	-	Density of Casting Powder (Mercury-Macro Method), Determination of
OD 27923	-	Nitroglycerin Determination Specific Gravity Method

(Copies of these documents are available from the Department of the Navy, Indian Head Division, NSWC IHEODTD, Code E14, Surface Systems Engineering Branch, 3703 Benson Road, Indian Head, MD 20640-5085, OFFICIAL BUSINESS, or by email request to ihdiv_standardization_team@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CHEMICAL PROPULSION INFORMATION ANALYSIS CENTER

SPIA/PP8, Part I	-	Nitrocellulose Base Propellants: Method of Determining the Tensile Properties of Solid Rocket Propellants.

(Application for copies should be addressed to The Johns Hopkins University, Energetics Research Group, 10630 Little Patuxent Parkway, Suite 202, Columbia, MD 21044-3286, or by email request to info@erg.jhu.edu.)

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification For Woven Wire Test Sieve Cloth And Test Sieves

(Copies of these document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description of manufacturing process. Before beginning casting powder manufacture, a detailed description of manufacturing processes and equipment proposed for use should be prepared and approved, as specified (see 6.2).

3.2 Materials. The materials used in the manufacture of the casting powder shall conform to the requirements of applicable specifications unless specific approval in writing, covering a departure therefrom, has been obtained from the procuring activity prior to manufacture.

3.2.1 Nitrocellulose. The nitrocellulose shall conform to MIL-DTL-244, Grade A, Type I, except that the nitrocellulose shall be made from cotton linters.

3.2.2 Nitroglycerin. The nitroglycerin shall conform to MIL-N-246, Type I.

3.2.3 2-Nitrodiphenylamine. The 2-nitrodiphenylamine shall be in flake form and conform to MIL-N-3399. The purity shall be determined by the titanous chloride buffer method (4.3.2.1) or the bromination method (4.3.2.2) of MIL-N-3399.

3.2.4 Lead salicylate. The lead salicylate shall conform to the following requirements:

3.2.4.1 Form. Soft crystals.

3.2.4.2 Color. White.

3.2.4.3 Lead content. 43.0 percent, minimum.

3.2.4.4 Salicylate content. 55.4 percent, minimum.

3.2.4.5 Volatile matter. 0.25 percent, maximum.

3.2.4.6 Granulation. Not more than 0.1 percent of the lead salicylate shall be retained on a U.S. Standard No. 325 sieve.

3.2.4.7 Workmanship. The lead salicylate shall be free of foreign material such as dirt, grit, and other extraneous material.

3.2.5 Lead beta-resorcylate. The lead beta resorcylate shall conform to the following:

a. Total lead content on a dry basis shall be not greater than 41.3 or less than 39.7 percent.

b. Beta-resorcylic acid on a dry basis shall be not greater than 60.0 or less than 58.0 percent.

c. The drying loss shall be not greater than 3.00 percent.

3.2.6 Carbon black. The carbon black shall conform to Grade D of WS 23141.

3.2.7 Graphite. The graphite shall conform to MIL-G-155, Grade III or IV.

3.3 Casting powder composition (volatile-free basis). The casting powder composition shall conform to the requirements shown in table I.

3.4 Total lead content. The total lead content, derived from lead salicylate and lead beta-resorcylate added to casting powder composition, shall be not greater than 2.75 or less than 2.25 percent.

3.5 Total volatiles. The total volatiles shall be not greater than 1.00 percent.

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TABLE I. Casting powder compositions (volatile-free basis).

Type I			Type II	Type III	Type II & III
Ingredient	**Percent (by weight)**	**Permissible Variation**	**Percent (by weight)**	**Percent (by weight)**	**Permissible Variation**
Nitrocellulose	74.70	By difference	73.70	76.10	By difference
Nitroglycerine	17.00	±1.00	18.00	15.90	±1.00
2-Nitrodiphenylamine	2.00	±0.15	2.00	1.80	±0.20
Lead salicylate	3.00	±0.25	3.00	2.90	±0.30
Lead beta-resorcylate	3.00	±0.25	3.00	2.90	±0.30
Carbon black	0.30	+0.10, -0.00	0.30	0.40	±0.10
Graphite (added basis)	0.04	±0.01	0.04	0.04	±0.01

3.6 Density. The density shall be not less than 1.625 grams per cubic centimeter (g/cc).

3.7 Granule dimensions. The granule dimensions shall be an average length and diameter of 0.035±0.005 inch. The length-to-diameter (L/D) ratio of the granules shall be 1.00±0.08. The granules shall be non-perforated cylinders.

3.8 Chemical stability. The chemical stability shall be as follows:

a. At 100 minutes, the slope shall be less than 1.00 mm/min.

b. It shall take more than 100 minutes to reach 100 millimeters.

3.9 Screen loading density. The screen loading density shall be not less than 1.030 g/cc for Type I and shall be not less than 1.050 g/cc for Types II and III.

3.10 Heat of explosion. The heat of explosion shall be 930±20 calories per gram for each type.

3.11 Evaluation grains (test specimens).

3.11.1 Physical properties. The physical properties at 77±2 °F shall conform to the requirements shown in table II.

TABLE II. Physical properties.

	Type I	Types II & III
Physical property	**Minimum Value**	**Minimum Value**
Tensile strength	400 psi	350 psi
Elongation	40%	35%

3.11.2 Burning rate. The burning rate at 1000 pounds per square inch absolute (psia) pressure shall conform to the requirements shown in table III.

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TABLE III. Burning rate.

Temperature, ºF	Rate, Inches per Second	
-10	Not more than 0.06 below 70 °F rate	
70	Type I	Types II and III
	0.77±0.04	0.78±0.04
130	Not more than 0.05 above 70 °F rate	

3.12 Data. A test report with description sheets should be prepared in accordance with MIL-STD-1171, giving a history of ingredients and manufacture, including process control and summary sheets, and evaluation and processing data, as specified (see 6.2).

3.13 Workmanship. The casting powder shall be free of foreign material such as dirt, grit, and other extraneous material.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of material from one supplier manufactured under operating and processing conditions that shall make the final product a uniform blend, as specified (see 6.2).

4.3 Sampling.

4.3.1 Composite sampling. The minimum sample size shall be three containers. The containers selected shall be the primary samples representing the lot. A composite sample, of sufficient size to perform the tests required by this specification, shall be made by blending equal portions from each of the selected containers in a blending barrel for a minimum of 30 minutes. The composite sample shall be placed in airtight containers and labeled with powder designation, manufacturer, plant, contract number, lot number, and lot size. If it becomes apparent during sampling that the lot is not uniform, primary samples may be required to be tested for conformance to this specification. All primary samples shall be retained for further testing should the composite sample fail to meet requirements, as specified (see 6.2).

4.3.2 Lead salicylate sampling. Ten percent, but in no case more than 10 nor fewer than 3, of the containers, shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer, plant, contract or purchase order number, and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, primary samples may be required to be tested for conformance to this specification. All primary samples shall be retained for further testing should the composite sample fail to meet the requirements, as specified (see 6.2).

4.4 Conformance tests. The following conformance tests shall be performed on each lot. Failure of any sample to conform to any requirement of this specification shall be cause for rejection of the lot. Where two test methods are referenced, the first one listed shall take precedence. The percentage of all ingredients shall be calculated on a volatile-free basis.

4.4.1 Nitrocellulose. The nitrocellulose shall be determined in accordance with MIL-STD-286, Method 209.2.

4.4.2 Nitroglycerin. The nitroglycerin shall be determined in accordance with MIL-STD-286, Method 208.1, or by High Performance Liquid Chromatography (HPLC), Method 208.3.1, except that acetonitrile may be substituted for methylene chloride and the sample may be introduced directly to the solvent without extraction.

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4.4.3 2-Nitrodiphenylamine. The 2-nitrodiphenylamine shall be determined in accordance with MIL-STD-286, Method 218.4, or by High Performance Liquid Chromatography (HPLC), Method 208.3.1, except that acetonitrile may be substituted for methylene chloride and the sample may be introduced directly to the solvent without extraction.

4.4.4 Lead salicylate.

4.4.4.1 Inspection. Inspection shall be as specified in MIL-G-2550.

4.4.4.2 Determination of lead content.

a. Procedure.

(1) Transfer an accurately weighed portion of approximately 1 gram of the sample to a 150 milliliter beaker.

(2) Add 10 milliliters of distilled water, and then add slowly, with stirring, 20 milliliters of solution consisting of equal parts by volume of concentrated nitric acid and concentrated sulfuric acid.

(3) Place the beaker and contents on a hot plate and heat until dense white fumes of sulfuric trioxide are evolved.

(4) Allow the beaker to cool for about 2 minutes, add 5 milliliters of concentrated nitric acid, and heat again until dense fumes of sulfur trioxide are evolved.

(5) Repeat the addition of nitric acid and heating to fumes until all the organic matter is destroyed and the sulfuric acid solution is practically colorless.

(6) Remove the beaker from the hot plate and allow it to cool.

(7) Wash down the side walls of the beaker with about 10 milliliters of distilled water, and again heat to fumes of sulfur trioxide.

(8) Allow the beaker and contents to cool, then cautiously dilute to about 100 milliliters with distilled water.

(9) Again allow the mixture to cool, add 25 milliliters of ethyl alcohol, and let stand for at least 2 hours.

(10) Filter the precipitate of lead sulfate, using a tared gooch type crucible.

(11) Transfer the precipitate to the crucible with a stream of 1:20 sulfuric acid solution directed from a wash bottle.

(12) Wash the precipitate three times with the sulfuric acid solution and twice with 95 percent ethyl alcohol.

(13) Aspirate the crucible and contents until the odor of alcohol is no longer detectable with the vacuum line turned off.

(14) Dry in an oven for 30 minutes at 100 to 105 °C, ignite in a muffle furnace at 550 to 600 °C for 15 minutes, cool in a desiccator, and weigh.

b. Calculation. Calculate the gain in weight of the crucible to percent lead in the sample as follows:

$$Percent\ lead = \frac{68.32A}{W}$$

Where:

A = gm of ignited residue

W = gm of sample

4.4.4.3 Determination of salicylate content.

a. Procedure.

(1) Transfer an accurately weighed portion of approximately 0.20 gram of the sample to a 500 milliliter iodine flask.

(2) Dissolve the sample in 10 milliliter of 10 percent sodium hydroxide solution, and dilute to approximately 150 milliliter with distilled water.

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(3) Add 50 milliliters of 0.2 Normal bromide-bromate solution (5.6 grams $KBrO_3$ and 15 grams KBr per liter), measured accurately from a pipette, to the contents of the flask.

(4) Add 25 milliliters of 1:1 hydrochloric acid solution, stopper the flask, and pour a few milliliters of 10 percent potassium iodide solution around the stopper.

(5) Allow to stand for 30 minutes, carefully loosen the stopper, and add 25 milliliter of 10 percent potassium iodide solution to the contents of the flask by pouring it around the loosened stopper.

(6) Titrate the liberated iodine with standard 0.2 Normal sodium thiosulfate solution, adding starch as an indicator near the end of the titration.

(7) When close to the end point, stopper the flask and shake vigorously after the addition of each drop of titrant.

(8) At the end point, the suspension should be bright yellow in color.

(9) Conduct a blank determination using 50 milliliter of the 0.2 Normal bromide-bromate solution exactly as described above, except that the sample will be omitted.

b. Calculation. Calculate the salicylate content as follows:

$$Percent\ salicylate = \frac{(A - B)\ \times N \times 2.285}{W}$$

Where:

A = milliliter of sodium thiosulfate solution used for titration of blank

B = milliliter of sodium thiosulfate solution used for titration of sample

N = normality of sodium thiosulfate solution

W = grams of sample

4.4.4.4 Determination of volatile matter.

a. Procedure.

(1) Transfer an accurately weighed portion of approximately 5 grams of the sample to a tared glass stoppered weighing bottle approximately 60 millimeter x 30 millimeter.

(2) Dry in an oven with the stopper off for 2 hours at 100 to 105 °C, replace the stopper, cool in a desiccator, and weigh.

b. Calculation. Calculate the percent volatile matter as follows:

$$Percent\ volatile\ matter = \frac{A \times 100}{W}$$

Where:

A = loss in weight of sample

W = grams of sample

4.4.4.5 Determination of granulation.

a. Procedure.

(1) Transfer 50 grams of the sample to a 600 milliliter beaker and add approximately 300 milliliter of 1 percent aerosol solution previously saturated with lead salicylate.

(2) Stir until the sample is thoroughly wetted, then pour the suspension on a No. 325 U.S. Standard sieve complying with ASTM E11.

(3) Transfer any material adhering to the walls of the beaker to the sieve, using a stream of the aerosol solution directed from a wash bottle.

(4) With the aid of a soft brush and a stream of the aerosol solution, agitate the sample until no more material passes through the sieve.

(5) Wash any material retained on the sieve with distilled water, saturated with lead salicylate, and dry the sieve and contents in an oven for two hours at 100 to 105 °C.

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(6) Transfer the residue to a tared weighing dish, cool, and weigh.

b. Calculation. Calculate the percent retained on the sieve.

4.4.4.6 Resubmission and retest. If the composite sample or any primary sample subjected to test fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance. The resubmitted lot shall be accepted, provided that the new samples, selected in accordance with 4.3.2, pass all the tests required by 4.4.4, and that 3 additional primary samples pass any test or tests failed by the material on original submission.

4.4.5 Lead beta-resorcylate. The lead beta-resorcylate contents shall be determined in accordance with the following procedures:

4.4.5.1 Total lead content.

a. Procedure.

(1) Weigh a 0.7- to 1.0-gram sample to the nearest 0.0001 gram in a 400 milliliter beaker.

(2) Add 150 milliliter deionized water and 5 milliliter acetic acid. Stir to dissolve. The solution will be hazy at this point.

(3) Adjust the pH to 5.0 to 5.5 with sodium hydroxide. This will cause the solution to clear.

(4) Add 4 drops of xylenol orange indicator and titrate with 0.05 Normal EDTA to the yellow endpoint.

b. Calculation.

$$\frac{mL\ EDTA\ x\ N\ EDTA\ x\ 20.719}{sample\ weight} = \%Pb$$

4.4.5.2 Total volatiles. The drying loss when heating a 2-gram sample for 2 hours at 100±2 °C shall not exceed 3.0 percent.

4.4.5.3 Beta-resorcylic acid content.

a. Procedure.

(1) In separate 150- milliliter beakers, accurately weigh the following:

(a) 0.04 gram beta-resorcylic acid.

(b) 0.08 gram beta-resorcylic acid.

(c) 0.15 to 0.20 gram sample.

(2) Add 50 milliliter of hot deionized water (80 to 90 °C) and 2.0 milliliter of hydrochloric acid to each beaker.

(3) Stir until the sample and standards are cool and dissolved, with no insoluble material present.

(4) Quantitatively transfer the contents of each beaker into separate 250- milliliter volumetric flasks, using deionized water to aid in transfer.

(5) Completely dilute each flask with deionized water and mix well.

(6) Pipet a 3.0- milliliter portion of each solution into separate 100- milliliter volumetric flasks and again completely dilute with deionized water and mix well.

b. Procedure.

(1) Determine the absorbance of the diluted standards and sample in the 1 centimeter cell on the ultraviolet spectrophotometer set at 254 nanometers, using deionized water as a blank.

(2) Record the absorbance for each sample and standard.

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c. Calculation.

(1) Calculate the absorbance per gram (APG) for each standard and sample. Dilutions do not need to be considered, since samples and standards are diluted alike.

$$APG = \frac{Absorbance}{Weight\ (g)}$$

(2) Record the APG for each sample and standard.

(3) Calculate the average APG (BRA) (AAPG (BRA)) using the APG for both standards.

(4) Record the AAPG (BRA).

(5) Calculate the percent Beta-resorcylic acid as follows: Use percent volatiles (from previous calculation) for the dry basis calculation.

$$\%BRA = \frac{100\ x\ APG\ sample}{AAPG\ (BRA)}$$

$$Dry\ basis\ \%BRA = \frac{\%BRA}{(100\% - \%volatiles)/100}$$

Note: Calculate percent Beta-resorcylic acid on a dry basis to the hundredths place.

4.4.6 Carbon black and graphite. The carbon black and graphite contents shall be determined in accordance with the following procedure:

a. Procedure.

(1) Weigh a 3-gram sample to ±0.05 milligrams. Transfer to a 400- milliliter beaker.

(2) Add 50 milliliter of concentrated nitric acid to the beaker and sample, and place on hot plate. Heat solution about 45 minutes or until sample is completely decomposed.

(3) Remove beaker from hot plate, allow beaker to cool, add 50 milliliters of water, and reheat. Continue heating until carbon black and graphite coagulate in bottom of beaker.

(4) Filter through a tared fine Selas or equal crucible (Weight A). Wash with 2/1 ether/acetone solution.

(5) Place crucible in 100 °C drying oven for 2 hours. Allow crucible to cool and reweigh the crucible (Weight B). Weight gain (Weight B-A) is the total carbon black and graphite in the sample.

(6) Place crucible on filtering apparatus and fill crucible with 20 percent sodium hydroxide solution. Turn vacuum on and the carbon black will flow through the crucible. Rinse crucible twice with additional sodium hydroxide solution and then with water.

(7) Place crucible in 100 °C drying oven for 2 hours. Allow crucible to cool and reweigh (Weight C).

b. Calculation.

$$\%\ Carbon\ Black = \frac{Weight(B) - Weight(A)}{3.0(Sample\ Weight)} \times 100 - \%\ Graphite$$

$$\%\ Graphite = \frac{Weight(C) - Weight(A)}{3.0(Sample\ Weight)} \times 100$$

4.4.7 Total lead content. The total lead content shall be determined in accordance with MIL-STD-286, Method 311.1. The equivalence factor shall be 0.6833.

4.4.8 Total volatiles. The total volatiles shall be determined in accordance with MIL-STD-286, Method 103.3 or Method 103.1.3.

4.4.9 Density. The density shall be determined in accordance with OD 19477, OD 26254, or MIL-STD-286, Method 510.3.1.

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4.4.10 Granule dimensions. The granule dimensions shall be determined in accordance with one of the following procedures:

a. Sample preparation.

(1) Coat approximately three quarters of one side of a microscope slide with a thin layer of light lubricating oil or petroleum jelly.

(2) Place a minimum of 20 granules, selected at random from a blended sample, on coated surface of microscope slide. Dry mounting is also acceptable.

b. Procedure.

(1) Use a Bausch and Lomb Illuminator, catalogue No. 42-63-58 or equal, or an optical comparator, mounted so that granule shadow will project upon a flat surface perpendicular to axis of light beam.

(2) Adjust projector for sharp focus and calibrate projector magnification with a wire of diameter known to ±0.0005 inch.

(3) Measure length and diameter of each granule to nearest 0.001 inch by measuring from knife-cut entrance to knife-cut entrance.

c. Alternate procedure.

(1) Use a Bausch and Lomb Model MA chemical microscope, or equal, equipped with a calibrated 100 division ocular micrometer and with incident lighting on sample, to measure length and diameter of granules to nearest 0.001 inch.

(2) When ends of granules are irregular, determine average length by measuring from knife-cut entrance to knife-cut entrance.

d. Calculation. Calculate and report average length and diameter dimensions of granules and L/D.

$$\frac{L}{D} = \frac{Average\ length}{Average\ diameter}$$

4.4.12 Chemical stability. The chemical stability test shall be determined in accordance with MIL-STD-286, Method 406.1, after the specimen has been desiccated for a minimum of 16 hours.

4.4.13 Screen loading density. The screen loading density shall be determined in accordance with OD 25216.

4.4.14 Heat of explosion. The heat of explosion shall be determined in accordance with MIL-STD-286, Method 802.1, except that pre-purified argon gas shall be used.

4.4.15 Evaluation grains (test specimens). Evaluation grains shall be manufactured in accordance with OD 24654 or OD 19555 (with exceptions). The following exceptions apply only to OD 19555:

a. Step 11.1 shall read, “Prepare a minimum of 75 pounds of casting solvent in accordance with OD 17118.”

b. Step 12.5 shall read, “Apply 45±2 psig of pressure to air cylinder by slowly opening the air pressure regulator.”

c. Step 12.6 shall read, “Apply 30±2 psig of pressure to solvent head by slowly opening the air pressure regulator.”

d. Step 12.7 shall read, “Allow cast grain to remain at ambient room temperature for a minimum of 72 hours.”

e. Step 12.16 shall read, “Cure cast grain at 140±10 °F for a minimum of 120 hours.”

4.4.15.1 Casting solvent composition. The casting solvent composition, on a moisture-free basis, shall conform to the requirements shown in table IV.

4.4.15.1.1 Moisture content. The moisture content shall be not greater than 0.10 percent when determined in accordance with MIL-STD-286, Method 101.5.

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4.4.15.2 Physical properties. Physical properties of propellant evaluation grains, manufactured in accordance with OD 24654 or OD 19555, shall be determined in accordance with SPIA/PP8, part I, at a temperature of 77±2 °F with a strain rate of 0.1 inch per minute per inch. The effective gage length for calculation purposes shall be 2.0 inches. Specimens shall be tested not later than 72 hours after final machining.

TABLE IV. Casting, solvent composition.

Ingredient	Specification	Percent by Weight	Test Method
Nitroglycerin	MIL-N-246, Type I	75.00±0.20	OD 27923
2-Nitrodiphenylamine	MIL-N-3399, Flake-form	1.00±0.10	MIL-STD-286, Method 218.4
Triacetin	MIL-T-301	24.00	By difference

4.4.15.3 Burning rate. The burning rate shall be determined in accordance with OD 19554 by firing 4 rounds each at -10, 70, and 130 °F and approximately 1000 psia.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order, as specified (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These casting powders are intended to be blended with MIL-P-82698 casting powder to produce MIL-P-82701 casting powder for use in the manufacture of solid propellants for MK 12 boosters for Terrier and Standard (ER) surface to air missiles. Type II casting powder is also blended with JCM-1915 casting powder to produce POU-2 casting powder for use in the manufacture of solid propellant for MK 75 gas generators for Tomahawk missiles. Since the items for which these casting powders are to be used were designed for military ordnance, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification.
b. Type of casting powder (see 1.2).
c. Data requirements (see 3.1 and 3.12).
d. Lot size (see 4.2).
e. Provisions for retest (see 4.3.1 and 4.3.2).
f. Packaging requirements (see 5.1).
g. Delivery location.

6.3 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous material" (DoD 4145.26M) are applicable and should be specified in the contract as required by the Defense Acquisition Regulation (DAR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

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6.4 Type similarity. The casting powder compositions Types I and II specified herein are similar to, but not identical with, ABL Type 917 casting powder composition for ARP propellant.

6.5 Packaging and marking.

6.5.1 Packaging. Packaging should be in containers conforming to Naval Ordnance Station, Indian Head drawing 5755. Electrical continuity between top and bottom chimes and between bottom liner and bottom chime should be subjected to 100 percent testing. In all tests, electrical resistance should be not greater than 5.0 ohms.

6.5.2 Marking. Containers should be marked in accordance with MIL-STD-129, Army Drawing 20-4-77, and 49 CFR 171-177. Permanent marking in accordance with MIL-STD-129 should be placed on three sides of each container. Marking should include, but not be limited to, the following:

a. Title, number, and revision letter of this specification.

b. Name of manufacturer and location.

c. Date of manufacture.

d. Lot number.

e. Contract number.

f. Type of casting powder material therein (see 1.2).

6.6 Supersession information. No official documentation exists for MIL-P-22504B(OS), therefore, this document has been created using MIL-P-22504B(OS) as a guide. References to MIL-P-22504B(OS) should be replaced by this document. MIL-DTL-32298 Type I material meets the requirements of material conforming to MIL-P-22504A(OS), dated 13 June 1967. MIL-DTL-32298 Type II material meets the requirements of material conforming to Naval Sea Systems Command purchase description WS 2531A, dated 25 March 1966.

6.7 Subject term (key word) listing.

Gas generators

MIL-P-82701 casting powder

MK 12 boosters

POU-2 casting powder

Solid propellants

Standard (ER)

Terrier

Tomahawk

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – OS
(Project 1376-2017-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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